

Product Visual Search: full Colab pipeline

This notebook runs the complete project pipeline for submission:

- ResNet18 supervised training and retrieval evaluation
- Frozen CLIP retrieval evaluation
- Lightly fine-tuned CLIP image encoder using `articleType` supervision

Expected Google Drive files:

- `MyDrive/product_visual_search/project.zip`
- `MyDrive/product_visual_search/fashion-product-images-small.zip`

1. Mount Google Drive

Before running the notebook, set Colab runtime to GPU.

```
from google.colab import drive
drive.mount('/content/drive')
```

Mounted at /content/drive

2. Configuration

```
from pathlib import Path

DRIVE_ROOT = Path('/content/drive/MyDrive/product_visual_search_v2')
PROJECT_ZIP = DRIVE_ROOT / 'project.zip'
DATA_ZIP = DRIVE_ROOT / 'fashion-product-images-small.zip'
WORKDIR = Path('/content/product_visual_search')

# Full submission run settings.
EPOCHS = 20
RESUME_TRAINING = False

# Use these switches if you need to rerun only one part.
RUN_RESNET = True
RUN_FROZEN_CLIP = True
RUN_CLIP_FINETUNE = True

RESNET_BATCH_SIZE = 64
CLIP_BATCH_SIZE = 16
NUM_WORKERS = 2
UNFREEZE_VISUAL_BLOCKS = 1

print('PROJECT_ZIP:', PROJECT_ZIP)
print('DATA_ZIP:', DATA_ZIP)
print('WORKDIR:', WORKDIR)

PROJECT_ZIP: /content/drive/MyDrive/product_visual_search_v2/project.zip
DATA_ZIP: /content/drive/MyDrive/product_visual_search_v2/fashion-product-images-small.zip
WORKDIR: /content/product_visual_search
```

3. Check GPU and uploaded files

```
import torch

print('cuda_available:', torch.cuda.is_available())
print('device:', torch.cuda.get_device_name(0) if torch.cuda.is_available() else 'CPU')
assert torch.cuda.is_available(), 'Please switch Colab runtime to GPU before full training.'
assert PROJECT_ZIP.exists(), f'Missing {PROJECT_ZIP}'
assert DATA_ZIP.exists(), f'Missing {DATA_ZIP}'
print('project.zip MB:', round(PROJECT_ZIP.stat().st_size / 1024 / 1024, 2))
print('dataset.zip MB:', round(DATA_ZIP.stat().st_size / 1024 / 1024, 2))

cuda_available: True
device: Tesla T4
project.zip MB: 2.63
dataset.zip MB: 565.16
```



4. Unpack project and dataset

```
import shutil
import zipfile

if WORKDIR.exists():
    shutil.rmtree(WORKDIR)
WORKDIR.mkdir(parents=True, exist_ok=True)

tmp_dir = Path('/content/project_unpack')
if tmp_dir.exists():
    shutil.rmtree(tmp_dir)
tmp_dir.mkdir(parents=True, exist_ok=True)

with zipfile.ZipFile(PROJECT_ZIP, 'r') as zf:
    zf.extractall(tmp_dir)

children = list(tmp_dir.iterdir())
source_root = children[0] if len(children) == 1 and children[0].is_dir() else tmp_dir
for item in source_root.iterdir():
    target = WORKDIR / item.name
    if target.exists():
        if target.is_dir():
            shutil.rmtree(target)
        else:
            target.unlink()
    shutil.move(str(item), str(target))

raw_dir = WORKDIR / 'data' / 'raw'
raw_dir.mkdir(parents=True, exist_ok=True)
with zipfile.ZipFile(DATA_ZIP, 'r') as zf:
    zf.extractall(raw_dir)

myntra_dir = raw_dir / 'myntradatast'
if (myntra_dir / 'styles.csv').exists():
    shutil.copy2(myntra_dir / 'styles.csv', raw_dir / 'styles.csv')
if (myntra_dir / 'images').exists():
    target_images = raw_dir / 'images'
    if target_images.exists():
        shutil.rmtree(target_images)
    shutil.copytree(myntra_dir / 'images', target_images)

print('project root:', sorted(p.name for p in WORKDIR.iterdir()))
print('prepare split exists:', (WORKDIR / 'scripts' / 'prepare_full_dataset_split.py').exists())
print('styles.csv exists:', (raw_dir / 'styles.csv').exists())
print('image_count:', len(list((raw_dir / 'images').glob('*.*jpg'))))
```

```
project root: ['.gitattributes', '.gitignore', 'GITHUB_UPLOAD_INSTRUCTIONS.md', 'MODEL_AND_DATA_FILES.md', 'README.md', 'app.py', 'colab
prepare split exists: True
styles.csv exists: True
image_count: 44441
```



5. Install dependencies

```
%cd /content/product_visual_search
!python --version
!pip install -r requirements.txt
```

Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas->r requirements)

Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests<3,>=2.27->streamlit-

Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests<3,>=2.27->streamlit->r re

Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests<3,>=2.27->streamlit->r re

Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch->r requiremen

Requirement already satisfied: pyyaml in /usr/local/lib/python3.12/dist-packages (from timm>=1.0.17->open_clip_torch->r requirements.

Requirement already satisfied: h11>=0.8 in /usr/local/lib/python3.12/dist-packages (from uvicorn>=0.30.0->streamlit->r requirements. t

Requirement already satisfied: wcwidth in /usr/local/lib/python3.12/dist-packages (from ftfy->open_clip_torch->r requirements. txt (li

Requirement already satisfied: hf-xet<2.0.0,>=1.4.3 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub->open_clip_torch-

Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub->open_clip_torch->r

Requirement already satisfied: typer in /usr/local/lib/python3.12/dist-packages (from huggingface-hub->open_clip_torch->r requirement

Requirement already satisfied: smmap<6,>=3.0.1 in /usr/local/lib/python3.12/dist-packages (from gitdb<5,>=4.0.1->gitpython!=3.1.19,<4,

Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub->open

Requirement already satisfied: attrs>=22.2.0 in /usr/local/lib/python3.12/dist-packages (from jsonschema>=3.0->altair!=5.4.0,!5.4.1,<

Requirement already satisfied: jsonschema-specifications>=2023.03.6 in /usr/local/lib/python3.12/dist-packages (from jsonschema>=3.0->

Requirement already satisfied: referencing>=0.28.4 in /usr/local/lib/python3.12/dist-packages (from jsonschema>=3.0->altair!=5.4.0,!5

Requirement already satisfied: rpds-py>=0.25.0 in /usr/local/lib/python3.12/dist-packages (from jsonschema>=3.0->altair!=5.4.0,!5.4.1

Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub->open_clip_t

Requirement already satisfied: rich>=12.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub->open_clip_torch->

Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub->open_clip

Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer->huggingface

Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer->huggingfa

Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0->rich>=12.3.0->typer-

Downloading streamlit-1.57.0-py3-none-any.whl (9.2 MB)

9.2/9.2 MB 83.0 MB/s eta 0:00:00

Downloading open_clip_torch-3.3.0-py3-none-any.whl (1.5 MB)

1.5/1.5 MB 79.3 MB/s eta 0:00:00

Downloading python-docx-1.2.0-py3-none-any.whl (252 kB)

253.0/253.0 kB 28.8 MB/s eta 0:00:00

Downloading pydeck-0.9.2-py2.py3-none-any.whl (11.3 MB)

11.3/11.3 MB 95.5 MB/s eta 0:00:00

Downloading ftfy-6.3.1-py3-none-any.whl (44 kB)

44.8/44.8 kB 5.4 MB/s eta 0:00:00

Installing collected packages: python-docx, ftfy, pydeck, streamlit, open_clip_torch

Successfully installed ftfy-6.3.1 open_clip_torch-3.3.0 pydeck-0.9.2 python-docx-1.2.0 streamlit-1.57.0

5b. Runtime helpers and package sanity check

```
import subprocess
import sys

def run_cmd(command):
    if command.startswith('python '):
        command = command.replace('python ', 'python -u ', 1)
    print(command, flush=True)
    process = subprocess.Popen(
        command,
        shell=True,
        text=True,
        stdout=subprocess.PIPE,
        stderr=subprocess.STDOUT,
        bufsize=1,
    )
    assert process.stdout is not None
    for line in process.stdout:
        print(line, end='', flush=True)
    return_code = process.wait()
    if return_code != 0:
        raise RuntimeError(f'Command failed with exit code {return_code}: {command}')
```

6. Prepare split

```
%cd /content/product_visual_search
run_cmd('python scripts/prepare_full_dataset_split.py --skip-image-verify')
```

```
/content/product_visual_search
python -u scripts/prepare_full_dataset_split.py --skip-image-verify
valid_image_count=44405
num_classes=132
train_size=31070
val_size=6666
test_size=6669
dropped_rare_classes_count=10
missing_image_metadata_rows=5
unreadable_images=0
images_per_class_top20=
  articleType  image_count
  Tshirts      7066
  Shirts       3215
```

Casual Shoes	2845
Watches	2542
Sports Shoes	2036
Kurtas	1844
Tops	1762
Handbags	1759
Heels	1323
Sunglasses	1073
Wallets	936
Flip Flops	914
Sandals	897
Briefs	849
Belts	813
Backpacks	724
Socks	686
Formal Shoes	637
Perfume and Body Mist	613
Jeans	608

```

min_images_per_class=3
max_images_per_class=7066
median_images_per_class=59.5
split_csv=/content/product_visual_search/outputs/splits/full_dataset_split.csv
class_mapping_json=/content/product_visual_search/outputs/splits/full_class_mapping.json
stats_csv=/content/product_visual_search/outputs/reports/full_dataset_stats.csv
dropped_rare_classes_csv=/content/product_visual_search/outputs/reports/dropped_rare_classes.csv

```

7. Ensure local CLIP checkpoint

The frozen CLIP scripts expect the Hugging Face safetensors file at the project-local model path.

```

from huggingface_hub import hf_hub_download

local_clip = WORKDIR / 'models' / 'huggingface' / 'timm_vit_base_patch32_clip_224_openai' / 'open_clip_model.safetensors'
local_clip.parent.mkdir(parents=True, exist_ok=True)
if not local_clip.exists():
    downloaded = hf_hub_download(
        repo_id='timm/vit_base_patch32_clip_224_openai',
        filename='open_clip_model.safetensors',
    )
    shutil.copy2(downloaded, local_clip)
print('local_clip:', local_clip)
print('exists:', local_clip.exists())
print('size MB:', round(local_clip.stat().st_size / 1024 / 1024, 2))

```

```

/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:93: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set it as secret in your Colab secrets, and then use `from huggingface_hub import hf_hub_download` to download the files.
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
warnings.warn(
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads
open_clip_model.safetensors: 100% 605M/605M [00:05<00:00, 484MB/s]
local_clip: /content/product_visual_search/models/huggingface/timm_vit_base_patch32_clip_224_openai/open_clip_model.safetensors
exists: True
size MB: 577.11

```

8. Train and evaluate ResNet18

```

if RUN_RESNET:
    resume = ' --resume' if RESUME_TRAINING else ''
    cmd = (
        f'python scripts/train_resnet18_full.py --epochs {EPOCHS} --batch-size {RESNET_BATCH_SIZE} '
        f'--num-workers {NUM_WORKERS} --patience {EPOCHS} {resume}'
    )
    run_cmd(cmd)
else:
    print('ResNet training skipped')

```

```

resnet val epoch 20/20: 28% ██████████ | 29/105 [00:04<00:10, 6.96it/s]
resnet val epoch 20/20: 30% ██████████ | 31/105 [00:04<00:10, 6.85it/s]
resnet val epoch 20/20: 31% ██████████ | 33/105 [00:04<00:10, 6.95it/s]
resnet val epoch 20/20: 33% ██████████ | 35/105 [00:04<00:09, 7.14it/s]
resnet val epoch 20/20: 35% ██████████ | 37/105 [00:05<00:09, 7.18it/s]
resnet val epoch 20/20: 37% ██████████ | 39/105 [00:05<00:08, 7.67it/s]
resnet val epoch 20/20: 39% ██████████ | 41/105 [00:05<00:09, 6.52it/s]
resnet val epoch 20/20: 41% ██████████ | 43/105 [00:06<00:09, 6.25it/s]
resnet val epoch 20/20: 43% ██████████ | 45/105 [00:06<00:09, 6.19it/s]
resnet val epoch 20/20: 45% ██████████ | 47/105 [00:06<00:10, 5.49it/s]
resnet val epoch 20/20: 47% ██████████ | 49/105 [00:07<00:11, 4.97it/s]
resnet val epoch 20/20: 49% ██████████ | 51/105 [00:07<00:10, 5.01it/s]
resnet val epoch 20/20: 50% ██████████ | 53/105 [00:08<00:10, 4.83it/s]
resnet val epoch 20/20: 51% ██████████ | 54/105 [00:08<00:09, 5.27it/s]
resnet val epoch 20/20: 52% ██████████ | 55/105 [00:08<00:10, 4.86it/s]
resnet val epoch 20/20: 54% ██████████ | 57/105 [00:09<00:10, 4.60it/s]
resnet val epoch 20/20: 55% ██████████ | 58/105 [00:09<00:09, 5.17it/s]
resnet val epoch 20/20: 56% ██████████ | 59/105 [00:09<00:10, 4.53it/s]
resnet val epoch 20/20: 58% ██████████ | 61/105 [00:09<00:08, 5.15it/s]
resnet val epoch 20/20: 59% ██████████ | 62/105 [00:09<00:07, 5.74it/s]
resnet val epoch 20/20: 60% ██████████ | 63/105 [00:10<00:08, 5.06it/s]
resnet val epoch 20/20: 62% ██████████ | 65/105 [00:10<00:07, 5.22it/s]
resnet val epoch 20/20: 64% ██████████ | 67/105 [00:10<00:06, 5.85it/s]
resnet val epoch 20/20: 66% ██████████ | 69/105 [00:11<00:05, 6.72it/s]
resnet val epoch 20/20: 68% ██████████ | 71/105 [00:11<00:05, 6.65it/s]
resnet val epoch 20/20: 70% ██████████ | 73/105 [00:11<00:04, 7.38it/s]
resnet val epoch 20/20: 71% ██████████ | 75/105 [00:11<00:03, 7.56it/s]
resnet val epoch 20/20: 73% ██████████ | 77/105 [00:12<00:03, 7.47it/s]
resnet val epoch 20/20: 75% ██████████ | 79/105 [00:12<00:03, 7.52it/s]
resnet val epoch 20/20: 77% ██████████ | 81/105 [00:12<00:03, 7.53it/s]
resnet val epoch 20/20: 79% ██████████ | 83/105 [00:12<00:02, 7.73it/s]
resnet val epoch 20/20: 81% ██████████ | 85/105 [00:13<00:02, 8.18it/s]
resnet val epoch 20/20: 83% ██████████ | 87/105 [00:13<00:02, 8.34it/s]
resnet val epoch 20/20: 85% ██████████ | 89/105 [00:13<00:02, 7.85it/s]
resnet val epoch 20/20: 87% ██████████ | 91/105 [00:13<00:01, 7.71it/s]
resnet val epoch 20/20: 89% ██████████ | 93/105 [00:14<00:01, 7.68it/s]
resnet val epoch 20/20: 90% ██████████ | 95/105 [00:14<00:01, 7.64it/s]
resnet val epoch 20/20: 92% ██████████ | 97/105 [00:14<00:01, 7.99it/s]
resnet val epoch 20/20: 94% ██████████ | 99/105 [00:14<00:00, 8.05it/s]
resnet val epoch 20/20: 96% ██████████ | 101/105 [00:15<00:00, 8.40it/s]
resnet val epoch 20/20: 97% ██████████ | 102/105 [00:15<00:00, 8.24it/s]
resnet val epoch 20/20: 98% ██████████ | 103/105 [00:15<00:00, 8.21it/s]
resnet val epoch 20/20: 100% ██████████ | 105/105 [00:15<00:00, 6.76it/s]
periodic_checkpoint=/content/product_visual_search/outputs/checkpoints/resnet18_full_epoch_020.pth
epoch=20 train_loss=0.0220 val_loss=0.5149 val_top1=0.8918 val_top5=0.9889 epoch_time=147.3s lr=0.0001
last_checkpoint=/content/product_visual_search/outputs/checkpoints/resnet18_full_last.pth
best_checkpoint=/content/product_visual_search/outputs/checkpoints/resnet18_full_best.pth
training_log=/content/product_visual_search/outputs/reports/resnet18_full_training_log.csv

```

```

if RUN_RESNET:
    run_cmd(f'python scripts/evaluate_resnet18_full.py --batch-size {RESNET_BATCH_SIZE} --num-workers {NUM_WORKERS}')
    run_cmd(f'python scripts/build_resnet18_full_index.py --batch-size {RESNET_BATCH_SIZE} --num-workers {NUM_WORKERS}')
    run_cmd(f'python scripts/evaluate_resnet18_full_retrieval.py --batch-size {RESNET_BATCH_SIZE} --num-workers {NUM_WORKERS}')
else:
    print('ResNet evaluation skipped')

```

```

python -u scripts/evaluate_resnet18_full.py --batch-size 64 --num-workers 2
{
    "test_loss": 0.41876709798158496,
    "test_top1_acc": 0.8947368421052632,
    "test_top5_acc": 0.9880041985305144,
    "test_size": 6669,
    "train_size": 31070,
    "val_size": 6666,
    "num_classes": 132,
    "checkpoint": "/content/product_visual_search/outputs/checkpoints/resnet18_full_best.pth"
}
metrics_json=/content/product_visual_search/outputs/reports/resnet18_full_test_metrics.json
per_class_accuracy_csv=/content/product_visual_search/outputs/reports/resnet18_full_per_class_accuracy.csv
summary_csv_updated=/content/product_visual_search/outputs/reports/experiment_summary.csv
python -u scripts/build_resnet18_full_index.py --batch-size 64 --num-workers 2
gallery_size=31070
embedding_shape=(31070, 512)
index_scope=full train gallery
output_path=/content/product_visual_search/outputs/indexes/resnet18_full_gallery_index.npz
python -u scripts/evaluate_resnet18_full_retrieval.py --batch-size 64 --num-workers 2
{
    "recall@1": 0.9016344279502174,
    "recall@5": 0.9392712550607287,
    "recall@10": 0.9497675813465287,
    "query_size": 6669,
    "gallery_size": 31070,
    "embedding_dim": 512,
    "index_path": "/content/product_visual_search/outputs/indexes/resnet18_full_gallery_index.npz",
    "is_smoke_test": false
}
metrics_json=/content/product_visual_search/outputs/reports/resnet18_full_retrieval_metrics.json

```

```

resnet_backup = DRIVE_ROOT / 'outputs_after_resnet'
resnet_backup.mkdir(parents=True, exist_ok=True)
!cp -r /content/product_visual_search/outputs {resnet_backup}
print('Backed up ResNet stage to', resnet_backup)

```

Backed up ResNet stage to /content/drive/MyDrive/product_visual_search_v2/outputs_after_resnet

9. Evaluate frozen CLIP

```

if RUN_FROZEN_CLIP:
    run_cmd('python scripts/build_clip_full_index.py')
    run_cmd(f'python scripts/evaluate_clip_full_retrieval.py --batch-size {CLIP_BATCH_SIZE}')
else:
    print('Frozen CLIP skipped')

```

```

python -u scripts/build_clip_full_index.py
checkpoint_source=/content/product_visual_search/models/huggingface/timm_vit_base_patch32_clip_224_openai/open_clip_model.safetensors
download_new_model=False
index_path=/content/product_visual_search/outputs/indexes/clip_full_gallery_index.npz
gallery_size=31070
embedding_shape=(31070, 512)
query_size=6669
num_classes=132
label_column=articleType
index_scope=full train gallery
python -u scripts/evaluate_clip_full_retrieval.py --batch-size 16
{
    "recall@1": 0.8289098815414605,
    "recall@5": 0.9542660068975859,
    "recall@10": 0.9725596041385515,
    "query_size": 6669,
    "gallery_size": 31070,
    "num_classes": 132,
    "embedding_dim": 512,
    "embedding_extraction_time_seconds": 34.789523642999484,
    "checkpoint_source": "/content/product_visual_search/models/huggingface/timm_vit_base_patch32_clip_224_openai/open_clip_model.safetensors",
    "index_path": "/content/product_visual_search/outputs/indexes/clip_full_gallery_index.npz",
    "is_smoke_test": false
}
metrics_json=/content/product_visual_search/outputs/reports/clip_full_retrieval_metrics.json

```

10. Train and evaluate fine-tuned CLIP

```

if RUN_CLIP_FINETUNE:
    resume = ' --resume' if RESUME_TRAINING else ''
    cmd = (
        f"python scripts/train_clip_articletype.py --epochs {EPOCHS} --batch-size {CLIP_BATCH_SIZE} "
        f"--num-workers {NUM_WORKERS} --unfreeze-visual-blocks {UNFREEZE_VISUAL_BLOCKS} "
        f"--patience {EPOCHS} {resume}"
    )
    run_cmd(cmd)
else:
    print('Fine-tuned CLIP training skipped')

```

```

clip-ft val epoch 20/20: 88% ██████████ | 367/417 [00:24<00:03, 14.39it/s]
clip-ft val epoch 20/20: 88% ██████████ | 369/417 [00:24<00:03, 14.79it/s]
clip-ft val epoch 20/20: 89% ██████████ | 371/417 [00:24<00:03, 14.83it/s]
clip-ft val epoch 20/20: 89% ██████████ | 373/417 [00:24<00:02, 14.70it/s]
clip-ft val epoch 20/20: 90% ██████████ | 375/417 [00:24<00:02, 14.92it/s]
clip-ft val epoch 20/20: 90% ██████████ | 377/417 [00:24<00:02, 14.48it/s]
clip-ft val epoch 20/20: 91% ██████████ | 379/417 [00:25<00:02, 14.33it/s]
clip-ft val epoch 20/20: 91% ██████████ | 381/417 [00:25<00:02, 14.62it/s]
clip-ft val epoch 20/20: 92% ██████████ | 383/417 [00:25<00:02, 14.40it/s]
clip-ft val epoch 20/20: 92% ██████████ | 385/417 [00:25<00:02, 14.81it/s]
clip-ft val epoch 20/20: 93% ██████████ | 387/417 [00:25<00:02, 14.80it/s]
clip-ft val epoch 20/20: 93% ██████████ | 389/417 [00:25<00:01, 14.66it/s]
clip-ft val epoch 20/20: 94% ██████████ | 391/417 [00:25<00:01, 14.80it/s]
clip-ft val epoch 20/20: 94% ██████████ | 393/417 [00:25<00:01, 14.84it/s]
clip-ft val epoch 20/20: 95% ██████████ | 395/417 [00:26<00:01, 14.87it/s]
clip-ft val epoch 20/20: 95% ██████████ | 397/417 [00:26<00:01, 14.70it/s]
clip-ft val epoch 20/20: 96% ██████████ | 399/417 [00:26<00:01, 14.74it/s]
clip-ft val epoch 20/20: 96% ██████████ | 401/417 [00:26<00:01, 14.24it/s]
clip-ft val epoch 20/20: 97% ██████████ | 403/417 [00:26<00:00, 14.40it/s]
clip-ft val epoch 20/20: 97% ██████████ | 405/417 [00:26<00:00, 14.37it/s]
clip-ft val epoch 20/20: 98% ██████████ | 407/417 [00:26<00:00, 14.53it/s]
clip-ft val epoch 20/20: 98% ██████████ | 409/417 [00:27<00:00, 14.52it/s]
clip-ft val epoch 20/20: 99% ██████████ | 411/417 [00:27<00:00, 14.77it/s]
clip-ft val epoch 20/20: 99% ██████████ | 413/417 [00:27<00:00, 14.72it/s]
clip-ft val epoch 20/20: 100% ██████████ | 415/417 [00:27<00:00, 15.11it/s]
clip-ft val epoch 20/20: 100% ██████████ | 417/417 [00:27<00:00, 16.25it/s]
clip-ft val epoch 20/20: 100% ██████████ | 417/417 [00:27<00:00, 15.08it/s]
periodic_checkpoint=/content/product_visual_search/outputs/checkpoints/clip_articletype_epoch_020.pth
epoch=20 train_loss=0.1838 val_loss=0.5708 val_top1=0.8615 val_top5=0.9842 epoch_time=180.1s lr=0.0005
last_checkpoint=/content/product_visual_search/outputs/checkpoints/clip_articletype_last.pth
best_checkpoint=/content/product_visual_search/outputs/checkpoints/clip_articletype_best.pth
training_log=/content/product_visual_search/outputs/reports/clip_articletype_training_log.csv

```

```

if RUN_CLIP_FINETUNE:
    run_cmd(f'python scripts/evaluate_clip_articletype.py --batch-size {CLIP_BATCH_SIZE} --num-workers {NUM_WORKERS}')
    run_cmd(f'python scripts/build_clip_articletype_index.py --batch-size {CLIP_BATCH_SIZE} --num-workers {NUM_WORKERS}')
    run_cmd(f'python scripts/evaluate_clip_articletype_retrieval.py --batch-size {CLIP_BATCH_SIZE} --num-workers {NUM_WORKERS}')
else:
    print('Fine-tuned CLIP evaluation skipped')

```

```

python -u scripts/evaluate_clip_articletype.py --batch-size 16 --num-workers 2
WARNING:root:No pretrained weights loaded for model 'ViT-B-32'. Model initialized randomly.
{
  "test_loss": 0.5033388058496462,
  "test_top1_acc": 0.8666966561703404,
  "test_top5_acc": 0.9847053531264057,
  "test_size": 6669,
  "train_size": 31070,
  "val_size": 6666,
  "num_classes": 132,
  "checkpoint": "/content/product_visual_search/outputs/checkpoints/clip_articletype_best.pth"
}
metrics_json=/content/product_visual_search/outputs/reports/clip_articletype_test_metrics.json
per_class_accuracy_csv=/content/product_visual_search/outputs/reports/clip_articletype_per_class_accuracy.csv
summary_csv_updated=/content/product_visual_search/outputs/reports/experiment_summary.csv
python -u scripts/build_clip_articletype_index.py --batch-size 16 --num-workers 2
WARNING:root:No pretrained weights loaded for model 'ViT-B-32'. Model initialized randomly.
gallery_size=31070
embedding_shape=(31070, 512)
index_scope=full train gallery
output_path=/content/product_visual_search/outputs/indexes/clip_articletype_full_gallery_index.npz
python -u scripts/evaluate_clip_articletype_retrieval.py --batch-size 16 --num-workers 2
WARNING:root:No pretrained weights loaded for model 'ViT-B-32'. Model initialized randomly.
{
  "recall@1": 0.8612985455090718,
  "recall@5": 0.9412205727995202,
  "recall@10": 0.9574149047833258,
  "query_size": 6669,
  "gallery_size": 31070,
  "num_classes": 132,
  "embedding_dim": 512,
  "embedding_extraction_time_seconds": 29.468707657999403,
  "checkpoint_source": "/content/product_visual_search/models/huggingface/timm_vit_base_patch32_clip_224_openai/open_clip_model.safetensors",
  "index_path": "/content/product_visual_search/outputs/indexes/clip_articletype_full_gallery_index.npz",
  "is_smoke_test": false
}
metrics_json=/content/product_visual_search/outputs/reports/clip_articletype_retrieval_metrics.json

```

11. Show summary table

```

import pandas as pd

summary_path = WORKDIR / 'outputs' / 'reports' / 'experiment_summary.csv'
summary = pd.read_csv(summary_path)
summary[summary['experiment_mode'] == 'full']

```

```
summary_dataframe = experiment_metrics_dataframe[['model_name', 'top1_acc', 'top5_acc', 'recall@1', 'recall@5', 'recall@10',
'embedding_extraction_time_seconds', 'embedding_dim']]
```

	model_name	top1_acc	top5_acc	recall@1	recall@5	recall@10	embedding_extraction_time_seconds	embedding_dim
2	resnet18_full_dataset	0.894737	0.988004	0.901634	0.939271	0.949768		NaN
3	clip_vit_b_32_openai_full	NaN	NaN	0.828910	0.954266	0.972560	34.789524	
4	clip_vit_b_32_articletype_lite	NaN	NaN	0.861299	0.941221	0.957415	29.468708	

12. Back up final outputs

```
final_backup = DRIVE_ROOT / 'outputs_all_models_final'
if final_backup.exists():
    shutil.rmtree(final_backup)
final_backup.mkdir(parents=True, exist_ok=True)
!cp -r /content/product_visual_search/outputs {final_backup}
print('Final outputs saved to', final_backup)
```

Final outputs saved to /content/drive/MyDrive/product_visual_search_v2/outputs_all_models_final

13. Key files

```
key_files = [
    WORKDIR / 'outputs' / 'reports' / 'experiment_summary.csv',
    WORKDIR / 'outputs' / 'reports' / 'resnet18_full_test_metrics.json',
    WORKDIR / 'outputs' / 'reports' / 'resnet18_full_retrieval_metrics.json',
    WORKDIR / 'outputs' / 'reports' / 'clip_full_retrieval_metrics.json',
    WORKDIR / 'outputs' / 'reports' / 'clip_articletype_test_metrics.json',
    WORKDIR / 'outputs' / 'reports' / 'clip_articletype_retrieval_metrics.json',
    WORKDIR / 'outputs' / 'checkpoints' / 'resnet18_full_best.pth',
    WORKDIR / 'outputs' / 'checkpoints' / 'clip_articletype_best.pth',
]
for path in key_files:
    print(path.name, '=>', path.exists())
```

```
experiment_summary.csv => True
resnet18_full_test_metrics.json => True
resnet18_full_retrieval_metrics.json => True
clip_full_retrieval_metrics.json => True
clip_articletype_test_metrics.json => True
clip_articletype_retrieval_metrics.json => True
resnet18_full_best.pth => True
clip_articletype_best.pth => True
```